Roll No.: | | | | | | | | | | | | |

Test Date: 25-10-2026

Aakash
Medical | IIT-JEE | Foundations

D
Booklet Code

Aakash Delhi/NCR Zone



A I A T S
ALL INDIA AAKASH TEST SERIES

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for

Medical Entrance Exam - 2028

National Eligibility-cum-Entrance Test (NEET)

TEST No. 2

(XI Studying Students)

(Phase-1 & 2)

INSTRUCTIONS FOR CANDIDATES

1. Read each question carefully.
2. It is mandatory to use Blue/Black Ball Point Pen to darken the appropriate circle in the answer sheet.
3. Mark should be dark and should completely fill the circle.
4. Rough work must not be done on the answer sheet.
5. Do not use white-fluid or any other rubbing material on answer sheet. No change in the answer once marked is allowed.
6. Student cannot use log tables and calculators or any other material in the examination hall.
7. Before attempting the question paper, student should ensure that the test paper contains all pages and no page is missing.
8. Before handing over the answer sheet to the invigilator, candidate should check that Roll No. and Centre Code have been filled and marked correctly.
9. Immediately after the prescribed examination time is over, the answer sheet to be returned to the invigilator.
10. (i) The question paper consists of 180 compulsory questions (45 questions each in Physics and Chemistry and 90 questions in Biology (Botany & Zoology)). The total duration of the examination will be 180 minutes (3 Hrs). (ii) Each correct answer carries four marks. One mark will be deducted for each incorrect answer from the total score.

Note.: It is compulsory to fill **Roll No.** and **Test Booklet Code** on answer sheet, otherwise your answer sheet will not be considered.

Test No. 2

TOPICS OF THE TEST

Physics	Motion in a Plane
Chemistry	Classification of Elements and Periodicity in Properties, Chemical Bonding and Molecular Structure
Botany	Cell Cycle and Cell Division, The Living World
Zoology	Breathing and Exchange of Gases, Body Fluids and Circulation

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KARGOPOL

MM : 720

TEST - 2

Time : 3 Hrs.

[PHYSICS]

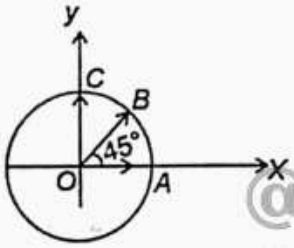
Choose the correct answer:

1. The coordinates of a particle moving in x-y plane is given by $x = a \sin(\pi t)$ and $y = b \cos(\pi t)$, where $a = b \neq 0$ are both constants while t is any time instant. The path of the particle will be

(1) Circular (2) Straight line

(3) Parabolic (4) Elliptical

2. The resultant vector of three vectors $\overline{OA}, \overline{OB}$ and $\overline{OC}$ that stem from the origin to the circumference of the circle (of radius r) which lies in x-y plane as shown in the figure, is



(1) $\left(\sqrt{2}r + r\right)$ along $\overline{OB}$ (2) $\left(\sqrt{2}r + r\right)$ along $\overline{BO}$

(3) $\left(\frac{r}{\sqrt{2}} + r\right)$ along $\overline{OB}$ (4) $\left(\frac{r}{\sqrt{2}} + r\right)$ along $\overline{BO}$

3. The position vector of an object at any time (t) is given by, $3t\hat{i} + 6t\hat{j} + 2\hat{k}$. Its velocity along y-axis has magnitude

(1) 3 unit (2) 6 unit

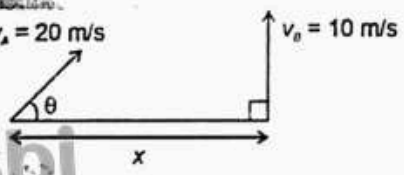
(3) 9 unit (4) Zero

4. A particle is projected from the origin in x-y plane under gravity, with y-axis along vertical and x-axis along the ground. The equation of its path is given by $6y = 2\sqrt{3}x - gx^2$, where x and y are in m. Then the speed of projection (in m/s) is

(1) 4 (2) 3

(3) 2 (4) 8

5. Two particles A and B are projected with velocity $v_A = 20$ m/s and $v_B = 10$ m/s from a smooth horizontal plane as shown in figure. If both particles collide after $\frac{1}{2}$ s, then value of x will be



(1) $5\sqrt{3}$ m (2) $4\sqrt{3}$ m

(3) $3\sqrt{3}$ m (4) $2\sqrt{3}$ m

6. If two vectors $\vec{a} = (2\hat{i} + 3\hat{j} - 2\hat{k})$ and $\vec{b} = (\hat{i} + \lambda\hat{j} + \mu\hat{k})$ are parallel to each other, then the value of $\left(\frac{\lambda}{\mu}\right)$ will be

(1) $-\frac{1}{2}$ (2) $-\frac{3}{2}$

(3) $\frac{1}{3}$ (4) $\frac{3}{4}$

Space for Rough Work

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7. A body starts from rest and covers a displacement $6\hat{i}+8\hat{j}$ in first 2 s, with uniform acceleration. The magnitude of acceleration of the body will be
(1) 4 m/s² (2) 6 m/s²
(3) 5 m/s² (4) 3 m/s²

8. Car *A* is moving with velocity 10 m/s 37° north of east, and a car *B* is moving with velocity 10 m/s 37° north of west. The velocity of car *A* w.r.t. car *B* will be
(1) 16 m/s toward east (2) 16 m/s toward west
(3) 12 m/s toward north (4) 12 m/s toward south

9. A displacement vector 5 km north is multiplied by –2 numerically, then the resultant vector becomes
(1) 10 km North (2) 10 km South
(3) 5 km North (4) 5 km South

10. Consider the following statements about the uniform circular motion and choose the correct option:
(a) Magnitude of change in velocity is zero
(b) Change in magnitude of velocity is zero
(c) Change in magnitude of acceleration is zero
(d) Magnitude of change in acceleration is zero
(1) Only (a) and (b) are correct
(2) Only (a) and (d) are correct
(3) Only (b) and (c) are correct
(4) Only (b) and (d) are correct

11. A vector of magnitude *R* makes an angle θ with the *x*-axis. The product of *x*-component and *y*-component of the vector will be
(1) $R^2\sin^2\theta$ (2) $R^2\sin2\theta$
(3) $\frac{R^2\sin2\theta}{2}$ (4) $\frac{R^2\sin^2\theta}{2}$

12. If $\vec{A}+\vec{B}=\vec{C}$ and angle between vectors $\vec{A}$ and $\vec{B}$ is 180°, then magnitude of $\vec{C}$ $\left(\text{assume } |\vec{A}|>|\vec{B}|\right)$
(1) Must be equal to $|\vec{A}|-|\vec{B}|$
(2) Must be less than $|\vec{A}|-|\vec{B}|$
(3) Must be greater than $|\vec{A}|+|\vec{B}|$
(4) Must be equal to $|\vec{A}|+|\vec{B}|$

13. An object is moving in *x*-*y* plane along the path $y=2x^2$, where *x* and *y* are in meter. If $x=t^2$, then *y*-component of velocity of object at time $t=0.5$ s will be
(1) 1 m/s (2) 4 m/s
(3) $\frac{1}{2}$ m/s (4) 2 m/s

14. An object is projected with speed 40 m/s at an angle 53° with the horizontal. When the velocity vector of the object makes an angle 37° with horizontal, then its vertical component of velocity will be
(1) 24 m/s (2) 12 m/s
(3) 18 m/s (4) 6 m/s

15. Consider the following statements and choose the correct option.
Statement (A): In oblique projectile motion, speed of the object continuously changes.
Statement (B): In oblique projectile motion, magnitude of acceleration of the object continuously changes.
(1) Both statement (A) and statement (B) are correct
(2) Both statement (A) and statement (B) are incorrect
(3) Statement (A) is correct but statement (B) is incorrect
(4) Statement (A) is incorrect but statement (B) is correct

Space for Rough Work

16. An object is projected in x - y plane with velocity $\vec{v} = (30\hat{i} + 40\hat{j})$ m/s, where x-axis is horizontal ground and y-axis is vertically upward (g = 10 m/s^2). Some parameters of the object are given in column-I while their values in SI unit are given in column-II. Match column-I with column-II and choose the correct option.

	Column-I		Column-II
(A)	Angle of projection with vertical	(P)	80
(B)	Maximum height	(Q)	37
(C)	Horizontal range	(R)	8
(D)	Time of flight	(S)	240

(1) A(P), B(S), C(Q), D(R)
(2) A(Q), B(P), C(S), D(R)
(3) A(R), B(S), C(P), D(Q)
(4) A(S), B(P), C(Q), D(R)

17. A boat is crossing the river with minimum drift. If speed of boat in still water is 8 km/h and speed of water is 4 km/h, then the angle made by steering of boat with river flow is

(1) 90° (2) 120°
(3) 150° (4) 0°

18. A vector parallel to the two-dimensional vector $(2\hat{i} + 2\hat{j})$ and having magnitude $4\sqrt{2}$ units, is

(1) $(8\hat{i} + 8\hat{j})$ (2) $(2\hat{i} + 2\hat{j})$
(3) $(4\hat{i} + 2\hat{j})$ (4) $(4\hat{i} + 4\hat{j})$

19. The magnitude of resultant of two vectors having magnitude $2p$ and $\sqrt{2}p$, is $\sqrt{10}p$. Then angle between both vectors will be

(1) 30° (2) 60°
(3) 45° (4) 90°

20. An object is moving on circular path with constant angular velocity in x-y plane. The direction of angular velocity must be

(1) Along positive z-direction
(2) Along negative z-direction
(3) In x-y plane
(4) Along positive or negative z-direction depending upon sense of rotation

21. Given below are two statements:

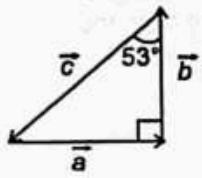
Statement A: If angle between velocity and acceleration (of an object) is obtuse, then magnitude of velocity decreases.

Statement B: If the angle between velocity and acceleration (of an object) remains 90°, then magnitude of velocity remains same.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both statement (A) and statement (B) are correct
(2) Both statement (A) and statement (B) are incorrect
(3) Statement (A) is correct and statement (B) is incorrect
(4) Statement (A) is incorrect and statement (B) is correct

22. In a right-angled triangle, the three vectors $\vec{a}, \vec{b}$ and $\vec{c}$ add up to zero as shown in figure. The angle between vectors $\vec{c}$ and $\vec{a}$ will be



(1) 127° (2) 53°
(3) 37° (4) 143°

Space for Rough Work

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23. An object is projected with velocity 10 m/s at an angle 60° with the horizontal ground. The ratio of radius of curvature at maximum height reached by the object to the maximum height itself, will be

(1) $\frac{3}{2}$ (2) 6

(3) 3 (4) $\frac{2}{3}$

24. An object is projected with velocity u at an angle 30° above the horizontal. The magnitude of displacement of the object when it will reach at maximum height, is

(1) $\frac{u^2\sqrt{21}}{8g}$ (2) $\frac{u^2\sqrt{13}}{8g}$

(3) $\frac{u^2\sqrt{13}}{4g}$ (4) $\frac{u^2\sqrt{21}}{4g}$

25. Two particles A and B are projected in x-y plane with speed u_1 and u_2 at angles θ_1 and θ_2 respectively with x-axis. Take x-axis on horizontal ground and y-axis vertically upward. If maximum height reached by both A and B are same, then (consider $\theta_1 > \theta_2$)

(1) Range of both A and B will be same

(2) Range of A is more than that of B

(3) Range of A is less than that of B

(4) Insufficient data

26. An object is projected in x-y plane with speed u = 10 m/s at an angle of 53° with x-axis. Take x-axis on horizontal ground and y-axis vertically upward. The rate of change of speed of the object at time t = 0 will be (g = 10 m/s²)

(1) 10 m/s² (2) 6 m/s²

(3) 8 m/s² (4) 4 m/s²

27. Which of the following is a vector?

(1) Displacement (2) Velocity

(3) Acceleration (4) All of these

28. Consider the following statements about ground-to-ground oblique projectile and choose the correct option:

(a) At highest point, speed of the object is minimum.

(b) Time taken to reach highest point is half of the total time of flight.

(c) Horizontal component of velocity continuously changes.

(1) Only (a) and (b) are correct

(2) Only (c) is correct

(3) Only (b) and (c) are correct

(4) All (a), (b) and (c) are correct

29. A particle is moving on circular path of radius R with constant angular speed π rad/s. The magnitude of displacement of the particle in a time interval of $\frac{2}{3}$ s will be

(1) $\frac{\sqrt{3}R}{2}$ (2) $2R$

(3) $\sqrt{2}R$ (4) $\sqrt{3}R$

30. In uniform circular motion, which among the following physical quantity remains constant?

(1) Velocity

(2) Angular velocity

(3) Centripetal acceleration

(4) Both (2) and (3)

31. A boat-man finds that he can save 5 s in crossing a river by quickest path, than by shortest path. If velocity of boat in still water and velocity of river are respectively 10 m/s and 8 m/s, then width of river will be

(1) 150 m (2) 75 m

(3) 100 m (4) 200 m

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Space for Rough Work

32. An object starts its motion from rest at the origin in x–y plane, with constant acceleration $\vec{a} = \left(2\hat{i} + 3\hat{j}\right)$ m/s². The equation of trajectory of the object will be

(1) $2y = 3x$ (2) $y = x$

(3) $y = \frac{3}{2}x^2$ (4) $y = \frac{2}{3}x^2$

33. For three vectors to be represented by the three sides of a triangle, their magnitudes cannot be

(1) 5, 9, 11 (2) 7, 10, 13

(3) 3, 8, 9 (4) 5, 7, 13

34. An object is projected horizontally with velocity v from certain height. Path of the object as seen by an observer moving upward with acceleration $a = 10$ m/s², will be (take $g = 10$ m/s²)

(1) Circular (2) Straight line

(3) Elliptical (4) Parabolic

35. Arrange the resultant magnitude (of vector addition) of two non-zero vectors $\vec{A}$ and $\vec{B}$ given below with different orientations, in decreasing order:

(a) $\vec{A}$ and $\vec{B}$ are parallel

(b) $\vec{A}$ and $\vec{B}$ are anti-parallel

(c) $\vec{A}$ and $\vec{B}$ are making an angle 60° with each other

(d) $\vec{A}$ and $\vec{B}$ are making an angle 120° with each other

(1) (a), (c), (d), (b) (2) (a), (d), (b), (c)

(3) (d), (b), (c), (a) (4) (d), (a), (b), (c)

36. Read the following statements and select the correct option.

Assertion (A): For a body projected from ground, speed of projection is equal to the speed of striking the ground.

Reason (R): Magnitude of horizontal component and vertical component of velocity are equal at the point of projection and the point of striking.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both (A) and (R) are correct and (R) is correct explanation of (A)

(2) Both (A) and (R) are correct but (R) is not correct explanation of (A)

(3) (A) is correct but (R) is incorrect

(4) (A) is incorrect but (R) is correct

37. A particle is projected from the ground in x-y plane with y-axis along vertical, the point of projection is origin. The equation of its path is given as $y = \sqrt{3}x - \frac{g}{2}x^2$, where y and x are in m. The time of flight of the particle will be (Take $g = 10$ m/s²)

(1) $\frac{\sqrt{3}}{4}$s (2) $\frac{\sqrt{3}}{5}$s

(3) $\frac{1}{4}$s (4) $\frac{1}{2}$s

38. Two projectiles A and B are thrown from the same point on ground with velocities v and $\frac{v}{2}$ respectively, such that B is thrown at an angle 45° with horizontal. Then the angle of projection of A with horizontal so that both projectiles have the same range, is

(1) $\sin^{-1}\left(\frac{1}{4}\right)$ (2) $\frac{1}{2}\sin^{-1}\left(\frac{1}{4}\right)$

(3) $2\sin^{-1}\left(\frac{1}{4}\right)$ (4) $\frac{1}{2}\sin^{-1}\left(\frac{1}{8}\right)$

Space for Rough Work

39. A particle is projected with velocity $\vec{v} = \left(10\hat{i} + 40\hat{j}\right)$ m/s from ground. The time after which the velocity vector makes an angle 45° with horizontal will be (take g = 10 m/s²)

(1) 4 s (2) 1 s

(3) 2 s (4) 3 s

40. A particle is at point A(0, 1, 3) m. If it starts moving from rest with a constant acceleration $\vec{a} = \left(2\hat{i} + 3\hat{k}\right)$ m/s², then the position of the particle at time t = 2 s will be

(1) $\left(2\hat{i} + 2\hat{j} + 3\hat{k}\right)$ m (2) $\left(\hat{i} + 2\hat{j} + 6\hat{k}\right)$ m

(3) $\left(4\hat{i} + \hat{j} + 9\hat{k}\right)$ m (4) $\left(3\hat{i} + \hat{j} + 9\hat{k}\right)$ m

41. Two ships A and B are 10 km apart on a line running south to north. Ship A (farther north) is streaming west at 20 km/h while ship B is streaming north at 20 km/h. The time after which both ships are closest to each other, is

(1) 30 min (2) 45 min

(3) 15 min (4) 60 min

42. A particle is moving in xyz-coordinate system from point A(1, 2, 1) to the point B(3, 5, 2). The angle made by displacement vector of the particle with x-axis will be

(1) $\theta = \cos^{-1}\left(\frac{4}{\sqrt{14}}\right)$ (2) $\theta = \cos^{-1}\left(\frac{2}{\sqrt{14}}\right)$

(3) $\theta = \cos^{-1}\left(\frac{3}{\sqrt{7}}\right)$ (4) $\theta = \cos^{-1}\left(\frac{1}{\sqrt{7}}\right)$

43. Given below are two statements:

Statement A: On rotating a vector about its tail, the rectangular components of the vector changes.

Statement B: A vector can be split into infinite number of components.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both statement (A) and statement (B) are correct

(2) Both statement (A) and statement (B) are incorrect

(3) Statement (A) is correct and statement (B) is incorrect

(4) Statement (A) is incorrect and statement (B) is correct

44. Rain is falling vertically with speed 6 km/h. A man is running with speed of 8 km/h on a horizontal plane. The angle with the vertical at which he must hold his umbrella to save himself from the rain, is

(1) 53°

(2) 30°

(3) 60°

(4) 37°

45. A body is projected from the ground at an angle θ with the horizontal. If the maximum height reached is h and time of flight is 6 s, then the value of h will be (take g = 10 m/s²)

(1) 20 m

(2) 60 m

(3) 45 m

(4) 90 m

Space for Rough Work

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[CHEMISTRY]

46. Consider the following species

(a) BCl_3 (b) BrF_3

(c) XeO_3 (d) NF_3

(e) $SbCl_5$

The species which contain 120° bond angle are

(1) (a), (b) and (c) only (2) (a) and (e) only

(3) (c), (d) and (e) only (4) (a), (b) and (d) only

47. An element (X) having atomic number 13 reacts with another element (Y) to give a compound X_2Y_3. The electronic configuration of Y could be

(1) $1s^2 2s^2 2p^3$ (2) $1s^2 2s^2 2p^4$

(3) $1s^2 2s^2 2p^5$ (4) $1s^2 2s^2 2p^6$

48. Given below are two statements, one is labelled as Assertion (A) and other is labelled as Reason (R):

Assertion (A): Ionisation of an atom is an endothermic process.

Reason (R) : Energy is always required to remove electron from an atom.

In the light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(2) Both (A) and (R) are true and (R) is the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

49. Which among the given sets represents isoelectronic species?

(1) F^-, Na^+, N^{3-} and Ca^{2+}

(2) Mn^{2+}, Fe^{3+}, Co^{3+} and Cr^+

(3) Cl^-, Ar, K^+ and Sc^{3+}

(4) Br^-, Rb^+, Sr^{2+} and Xe

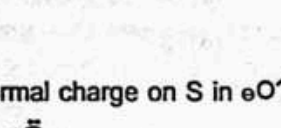
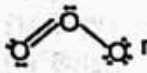
50. Correct order of covalent character of the given compounds is

(1) $AlCl_3 > MgCl_2 > NaCl$

(2) $MgCl_2 > NaCl > AlCl_3$

(3) $AlCl_3 > NaCl > MgCl_2$

(4) $NaCl > MgCl_2 > AlCl_3$

51. Formal charge on S in  and central O in  respectively are

(1) –2 and + 1 (2) +2 and +1

(3) Zero and +1 (4) Zero and Zero

52. Consider the following species

(a) NH_4Cl (b) $CaCl_2$

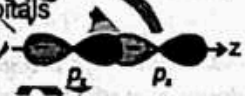
(c) $NaNO_3$ (d) $MgCl_2$

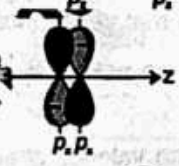
The species which contain both ionic and coordinate bonds is/are

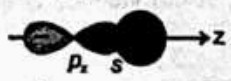
(1) (b) only (2) (a) and (c) only

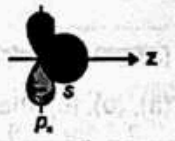
(3) (a), (b) and (c) only (4) (b) and (d) only

53. Consider the following overlapping of the atomic orbitals

(a) 

(b) 

(c) 

(d) 

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Space for Rough Work

The zero overlap is/are

(1) (a) and (b) only (2) (d) only

(3) (c) only (4) (b) and (d) only

54. Consider the following species

$[I_3]^+$, SO_2, $BeCl_2$, NO_2^+, H_2O, $[ICl_2]^-$, CO_2

The number of species which are linear in shape is

(1) 5 (2) 4

(3) 6 (4) 3

55. Consider the given statements

(a) In NO, the bond order is 2.5

(b) Bond order of NO^+ and N_2 are same

(c) Bond length of C_2 is greater than O_2

(d) Bond energy of F_2 is greater than B_2

The correct statements are

(1) (a), (b) and (c) only (2) (b) and (c) only

(3) (a) and (b) only (4) (a), (b), (c) and (d)

56. Consider the following statements about hybridisation

(a) The hybridised orbitals are always equivalent in energy and shape

(b) The number of hybrid orbitals is equal to the number of atomic orbitals that get hybridised

(c) Promotion of electron is an essential condition prior to hybridisation

(d) The orbitals present in the valence shell of the atom are hybridised

The correct statements are

(1) (a), (b) and (c) only (2) (a), (b) and (d) only

(3) (b) and (d) only (4) (a), (b), (c) and (d)

57. Match List-I with List-II

	List-I Elements		List-II Ionisation enthalpy ($kJ\ mol^{-1}$)
a.	F	(i)	1086
b.	C	(ii)	1402
c.	N	(iii)	1314
d.	O	(iv)	1681

The correct match is

(1) a(i), b(iv), c(ii), d(iii) (2) a(iv), b(i), c(ii), d(iii)

(3) a(iv), b(i), c(iii), d(ii) (4) a(ii), b(i), c(iv), d(iii)

58. The set which does not represent Dobereiner's triads is

(1) Ca, Sr, Ba (2) Cl, Br, I

(3) Li, Na, K (4) Zn, Cd, Hg

59. Consider the following statements

(a) General outer electronic configuration of inner transition elements is $(n-2)f^{1-14}(n-1)d^{0-1}ns^2$

(b) Selenium and Iodine are chalcogen and halogen respectively

(c) Bismuth and tellurium are semi-metals

(d) Vanadium and tin are main group elements

The correct statements are

(1) (a) and (b) only (2) (a), (b) and (c) only

(3) (c) and (d) only (4) (a), (b) and (d) only

60. The correct decreasing order of electronegativity of the given elements based on Pauling scale is

(1) Be > Mg > B > C (2) C > B > Be > Mg

(3) C > Be > B > Mg (4) B > C > Be > Mg

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61. Match List-I with List-II

	List-I Oxide		List-II Chemical Nature
a.	N_2O	(i)	Basic
b.	As_2O_3	(ii)	Neutral
c.	SO_2	(iii)	Amphoteric
d.	CaO	(iv)	Acidic

The correct match is

(1) a(iii), b(ii), c(iv), d(i) (2) a(ii), b(iii), c(i), d(iv)

(3) a(iv), b(iii), c(ii), d(i) (4) a(ii), b(iii), c(iv), d(i)

62. Given below are two statements, one is labelled as Assertion (A) and other is labelled as Reason (R):

Assertion (A): Methyl alcohol is soluble in water.

Reason (R): Methyl alcohol forms intermolecular hydrogen bond with water.

In the light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(2) Both (A) and (R) are true and (R) is the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

63. During change of B_2 to $B_2^{\oplus}$ ion, the electron adds to which of the following orbitals?

(1) σ orbital (2) σ^* orbital

(3) π orbital (4) π^* orbital

64. Shape of IF_5 and IF_7 respectively are

(1) Trigonal bipyramidal and pentagonal bipyramidal

(2) Trigonal bipyramidal and distorted octahedral

(3) Square pyramidal and pentagonal bipyramidal

(4) Square pyramidal and octahedral

65. The species which does not contain π bond is

(1) C_2H_4 (2) CO_2

(3) C_2H_2 (4) C_2H_6

66. In the structure of XeF_4 and BrF_3, the number of lone pair(s) of electrons on Xe and Br respectively are

(1) 3 and 1 (2) 2 and 2

(3) 2 and 3 (4) 4 and 2

67. Given below are the two statements

Statement I: A π bonding molecular orbital has larger electron density above and below the inter-nuclear axis.

Statement II: The π^* antibonding molecular orbital has a node between the nuclei.

In the light of above statements, choose the correct option.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

68. Consider the following pair of species

(a) NO_3^- and CO_3^{2-} (b) XeF_2 and IF_2^-

(c) ClO_4^- and PO_4^{3-} (d) NF_3 and ClF_3

Isostructural and isoelectronic pairs are

(1) (a) and (b) only (2) (b) and (c) only

(3) (a), (b) and (c) only (4) (a), (b), (c) and (d)

69. Correct order of atomic radii of the given elements is

(1) Si > P > S > C > N (2) P > Si > S > C > N

(3) Si > P > S > N > C (4) S > P > Si > N > C

70. Match List-I with List-II

	List-I Atomic Number		List-II IUPAC official name
a.	105	(i)	Rutherfordium
b.	108	(ii)	Dubnium
c.	102	(iii)	Nobelium
d.	104	(iv)	Hassium

The correct match is

(1) a(ii), b(iv), c(iii), d(i) (2) a(iv), b(ii), c(iii), d(i)

(3) a(ii), b(iv), c(i), d(iii) (4) a(iii), b(i), c(iv), d(ii)

Space for Rough Work

71. Given below are the two statements

Statement I: In OF_2 and Na_2O the oxidation state of O are + 2 and –2 respectively.

Statement II: Maximum covalency of boron in its compounds is 4.

In the light of above statements, choose the correct option.

(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

72. Consider the given statements

(a) Eka-aluminium is also known as Gallium
(b) According to modern periodic law, the properties of the elements are periodic function of their atomic weights
(c) Manganese belongs to seventh group of the periodic table
(d) Neptunium (Z = 93) is an element of lanthanoid series

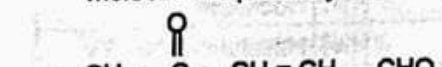
The correct statements are

(1) (a), (c) and (d) only (2) (a), (b) and (c) only
(3) (a) and (c) only (4) (c) and (d) only

73. The correct decreasing order of dipole moment of the given molecules is

(1) $NF_3 > CHCl_3 > NH_3 > CCl_4$
(2) $CCl_4 > NH_3 > CHCl_3 > NF_3$
(3) $NH_3 > CHCl_3 > NF_3 > CCl_4$
(4) $NH_3 > NF_3 > CHCl_3 > CCl_4$

74. The total number of σ bonds and π bonds in given molecule respectively are



(1) 12 and 2 (2) 5 and 3
(3) 12 and 3 (4) 11 and 3

75. Number of bonding electrons in N_2^+ and number of antibonding electrons in F_2 respectively are

(1) 8 and 7 (2) 7 and 9
(3) 9 and 6 (4) 9 and 8

76. If observed dipole moment of AB molecule is 1.5 D and A – B bond length is 120 pm then percentage ionic character in AB molecule is

(1) 45% (2) 26%
(3) 19% (4) 32%

77. Match List-I with List-II.

	List-I Bond type		List-II Covalent bond length (pm)
a.	C = O	(i)	138
b.	C = N	(ii)	96
c.	O – H	(iii)	107
d.	C – H	(iv)	121

The correct match is

(1) a(iv), b(i), c(iii), d(ii) (2) a(i), b(iv), c(ii), d(iii)
(3) a(ii), b(iii), c(i), d(iv) (4) a(iv), b(i), c(ii), d(iii)

78. Given below are the two statements.

Statement I: Octet theory is applicable to NO_2 but it is not applicable to SCl_2.

Statement II: Octet theory does not account for the shape of molecule..

In the light of above statements, choose the correct option.

(1) Statement I is correct but statement II is incorrect
(2) Statement I is incorrect but statement II is correct
(3) Both statement I and statement II are correct
(4) Both statement I and statement II are incorrect

Space for Rough Work

79. Consider the given species

(a) NF_3 (b) BF_3

(c) PCl_3 (d) H_2O

(e) $CHCl_3$

Non-planar and polar species are

(1) (a), (b) and (c) only (2) (a), (c) and (e) only

(3) (b) and (d) only (4) (c), (d) and (e) only

80. The correct decreasing bond order in the following species is

(1) $O_2^- > O_2^{2-} > O_2 > O_2^+$

(2) $O_2^+ > O_2 > O_2^- > O_2^{2-}$

(3) $O_2 > O_2^{2-} > O_2^- > O_2^+$

(4) $O_2^{2-} > O_2^- > O_2 > O_2^+$

81. Incorrect statement about PCl_5 molecule is

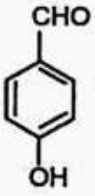
(1) Percentage p character in the hybrid orbitals of P is 60%

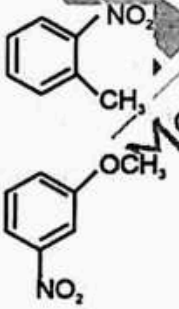
(2) Three P – Cl bonds lie in one plane and make an angle of 120° with each other

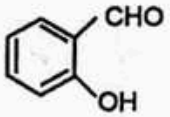
(3) Axial bonds are longer than equatorial bonds

(4) Maximum six atoms lie in a plane

82. Intramolecular hydrogen bond is present in

(1) 

(2) 

(3) 

(4)

83. Given below are the two statements.

Statement I: He_2 and Be_2 do not exist.

Statement II: B_2 contains one π bond and it is diamagnetic in nature.

In the light of above statements, choose the correct option.

(1) Statement I is correct but statement II is incorrect

(2) Both statement I and statement II are correct

(3) Both statement I and statement II are incorrect

(4) Statement I is incorrect but statement II is correct

84. Match List-I with List-II.

	List-I Molecule		List-II Hybridization of central atom
a.	SF_4	(i)	sp^3d^2
b.	XeF_4	(ii)	sp^3d
c.	XeO_3	(iii)	sp^2
d.	SO_2	(iv)	sp^3

The correct match is

(1) a(iv), b(ii), c(i), d(iii)

(2) a(ii), b(iv), c(iii), d(i)

(3) a(ii), b(i), c(iii), d(iv)

(4) a(iii), b(i), c(iv), d(iii)

85. The correct decreasing order of bond angles in the given species is

(1) $I_3^- > BCl_3 > CCl_4 > H_2O$

(2) $BCl_3 > I_3^- > H_2O > CCl_4$

(3) $BCl_3 > CCl_4 > I_3^- > H_2O$

(4) $I_3^- > CCl_4 > BCl_3 > H_2O$

Space for Rough Work

86. Consider the following species

(a) PO_4^{3-} (b) CO_3^{2-}

(c) SO_3 (d) ClO_4^-

$p\pi - d\pi$ bond(s) is/are present in

(1) (a) and (c) only (2) (b) and (d) only

(3) (a), (b) and (c) only (4) (a), (c) and (d) only

87. Element with atomic number 49 belongs to

(1) 5th period and 15th group

(2) 6th period and 14th group

(3) 5th period and 13th group

(4) 4th period and 15th group

88. IUPAC name of the elements with atomic number 101 and 109 respectively are

(1) Unununnium and Unnilunium

(2) Unnilennium and Unnilunium

(3) Unnilunium and Unnilennium

(4) Unununnium and Ununennium

89. Match List-I with List-II.

	List-I		List-II
a.	Vanadium	(i)	Non-metal
b.	Samarium	(ii)	Metalloid
c.	Germanium	(iii)	Transition metal
d.	Chlorine	(iv)	Lanthanoid

The correct match is

(1) a(iii), b(iv), c(i), d(ii) (2) a(iv), b(iii), c(ii), d(i)

(3) a(iv), b(iii), c(i), d(ii) (4) a(iii), b(iv), c(ii), d(i)

90. The correct decreasing order of negative electron gain enthalpy of the given elements is

(1) S > Se > Na > K (2) S > Se > K > Na

(3) Se > S > Na > K (4) Se > S > K > Na

[BOTANY]

91. The phase in which the homologous chromosomes of the bivalent exhibit the tendency to separate from each other also shows

(1) Formation of synaptonemal complex

(2) Decondensation of chromosomes

(3) The appearance of chiasmata

(4) Exchange of genetic material between homologous chromosomes

92. The most dramatic period of the cell cycle involves

(1) DNA and centriole duplication

(2) Cell growth due to protein synthesis

(3) Major reorganisation of virtually all components of the cell

(4) Formation of nuclear membrane during first phase of karyokinesis

93. Any sexually reproducing organism starts its life cycle from a

(1) Diploid cell

(2) Haploid cell

(3) Multinucleate cell

(4) Multicellular structure

94. Cell growth results in

(1) Disturbing the ratio between the nucleus and the cytoplasm

(2) Restoring the nucleo-cytoplasmic ratio

(3) Producing diploid daughter cells with identical genetic complement

(4) Producing haploid phase in the life cycle of sexually reproducing organisms

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95. According to binomial nomenclature rules, the biological names are generally in which of the following language?
(1) Greek (2) German
(3) Latin (4) English

96. Which of the following specific epithets does not belong to genus '*Solanum*'?
(1) *tuberosum* (2) *nigrum*
(3) *familiaris* (4) *melongena*

97. Tetrad of cells are obtained after cytokinesis that occurs
(1) At the end of telophase I
(2) Prior to the beginning of prophase II
(3) At the end of mitotic telophase
(4) At the end of telophase II

98. Read the following statements and select the **correct** option.
Statement A: In all organisms cell division stops with their maturity.
Statement B: Meiosis II is initiated just after the parental chromosomes have replicated to produce identical sister chromatids at the S-phase.
(1) Only statement A is correct
(2) Only statement B is correct
(3) Both statements A and B are correct
(4) Both statements A and B are incorrect

99. Which of the following information cannot be derived from the scientific name of the organism?
(1) Genus to which the organism belongs
(2) Author who proposed the name
(3) Name of the taxonomical category having lowest rank
(4) Habitat where the organism resides

100. Which of the following event does not takes place during interkinesis?
(1) Centriole duplication
(2) Synthesis of RNA
(3) Synthesis of proteins
(4) Replication of DNA

101. Match column-I with column-II and select the **correct** option.

	Column-I		Column-II
I.	Cell plate	a.	The structure contains four chromatids and two centromeres
II.	Bivalent	b.	This structural assembly forms during prophase
III.	Chiasmata	c.	This structure represents the middle lamella between the cells.
IV.	Mitotic apparatus	d.	This structure is seen after dissolution of synaptonemal complex

	I	II	III	IV
(1)	a	d	c	b
(2)	c	a	d	b
(3)	b	a	d	c
(4)	c	b	a	d

102. ICZN provides the agreed principles and criteria for the naming of organisms which
(1) Possess cellulosic cell wall
(2) Are eukaryotic
(3) Possess plastids
(4) Lack Golgi apparatus

Space for Rough Work

103. The correct scientific name of leopard is
(1) *Panthera leo* (2) *Panthera tigris*
(3) *Panthera pardus* (4) *Panthera aureus*

104. Disappearance of nucleolus and disintegration of nuclear membrane can be observed during
(1) Diakinesis (2) Diplotene
(3) Pachytene (4) Zygotene

105. Select the **incorrect** statement w.r.t. cell cycle.
(1) It occurs in a coordinated way to ensure formation of progeny cells with intact genomes.
(2) The events of cell cycle are under genetic control.
(3) The replicated chromosomes are distributed to daughter nuclei by a complex series of events during cell division.
(4) The period of cell cycle varies from organism to organism but always remains constant from cell to cell within an organism.

106. Carnivora includes families like
(1) Canidae and Hominidae
(2) Felidae and Muscidae
(3) Felidae and Canidae
(4) Hominidae and Muscidae

107. Families like Convolvulaceae and Solanaceae are placed in the
(1) Family Poales
(2) Class Sapindales
(3) Order Polymoniales
(4) Division Monocotyledonae

108. Crossing over takes place in
(1) Leptotene (2) Zygotene
(3) Pachytene (4) Diplotene

109. Read the below given **Assertion (A)** and **Reason (R)** statements:

Assertion (A): In telophase, the chromosomes that have reached their respective poles decondense and lose their individuality.

Reason (R): In telophase, the individual chromosomes can no longer be seen and each set of chromatin material tends to collect at each of the two poles.

In the light of the above statements, choose the **correct** answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

110. How many obligate taxonomic categories are shared by cat and gorilla?
(1) 4 (2) 3
(3) 5 (4) 2

111. Read the following statements and select the **correct** option.

Statement A: The taxonomic categories are distinct biological entities as well as the morphological aggregates.

Statement B: The basics of taxonomy like identification, naming and classification of organisms are universally evolved under international codes.
(1) Only statement A is correct
(2) Only statement B is correct
(3) Both statements A and B are correct
(4) Both statements A and B are incorrect

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Space for Rough Work

112. In animal cell, the cytokinesis is facilitated by
(1) Furrow formation
(2) Cell plate formation
(3) Phragmoplast formation
(4) Middle lamella formation

113. The cell that is in the G_0 phase of the cell cycle
(1) Arrests the cell division and does not allow it to continue ever again
(2) Is metabolically inactive phase but allows the synthesis of DNA
(3) Is the preparatory phase after which the cell can immediately perform karyokinesis
(4) Can enter into G_1 phase upon receiving the appropriate stimulus.

114. Which of the following pairs is **incorrectly** matched?

(1)	Yeast	–	Cell cycle is divided into two basic phases.
(2)	Heart cell	–	Usually do not exhibit the most dramatic phase of the cell cycle.
(3)	Onion root tip cell	–	Has both interphase and M phase.
(4)	Social insects	–	All of them shows mitosis in diploid cells only.

115. In the taxonomical hierarchy, kingdom as a taxonomical category
(1) Have the lowest rank
(2) Exhibit greatest difficulty of determining the relationship to other taxa at the same level
(3) Have the highest number of common characters in comparison to others
(4) Can be represented by 'Arthropoda'

116. Read the below given **Assertion (A)** and **Reason (R)** statements:

Assertion (A): Animals, 'Mammals' and 'Dogs' represent taxa at different levels.

Reason (R): Dog is a mammal and mammals are animals.

In the light of the above statements, choose the **correct** answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

117. All of the following are the contributions of Carolus Linnaeus, **except**
(1) Author of *Systema Naturae*
(2) Proposed the scientific name of mango
(3) Proposed a naming system which is being practiced by biologists all over the world
(4) Gave the most accepted definition of biological species

118. The complete disintegration of the nuclear envelope marks the start of
(1) Prophase (2) Metaphase
(3) Anaphase (4) Telophase

119. In mitosis, occurring in animal cell, the number of chromosomes in daughter cells depends on all of the following factors, **except**
(1) Splitting of centromere in anaphase
(2) Separation of chromatids in anaphase
(3) Proper attachment of chromosomes to the poles through spindle fibres
(4) Centrifugal manner of cytokinesis

Space for Rough Work

120. The sequence of events by which a cell duplicates its genome, synthesise the other constituents of the cell and eventually divided into two daughter cells is termed
(1) Cell duplication
(2) Cell death
(3) Cell cycle
(4) Cell division

121. The scientist who has been called 'The Darwin of the 20th century' is also
(1) Credited for proposing the molecular trafficking in the cell
(2) Awarded with the triple crown of biology
(3) Credited for enumerating number of species and describing the range in between 1.7–1.8 million
(4) Credited for proposing binomial nomenclature

122. Read the following statements of Assertion (A) and Reason (R)

Assertion (A): Gibbons are included in phylum chordata.

Reason (R) : Gibbons show the presence of a notochord and a dorsal hollow neural system.

In the light of above statements, select the correct option.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

123. Mitotic metaphase chromosomes are made up of
(1) Two sister chromatids held together by a centromere
(2) Two sister chromatids held together by two centromere
(3) Four sister chromatids held together by a centromere
(4) Four non-sister chromatids held together by a centromere

124. Meiosis II differs from mitosis as the former
(1) Shows metaphasic arrangement of chromosomes, in which sister chromatids are held together by primary constriction
(2) Occurs only in diploid cells
(3) Begins just after the duplication of DNA
(4) On completion produces daughter cells having different genetic composition than the parents

125. The least number of meiotic divisions required to form 100 pollen grains in a typical angiospermic plant is
(1) 50 (2) 25
(3) 100 (4) 125

126. Observe the below given taxons:
a. *Triticum* b. Monocotyledonae
c. Poaceae d. Angiospermae
e. Poales
Arrange these categories in the ascending order of taxonomical hierarchy.
(1) a → c → b → e → d (2) a → c → e → b → d
(3) a → d → c → b → e (4) b → e → a → c → d

127. Human cells in culture can progress through the cell cycle in only about
(1) 24 hours (2) 24 minutes
(3) 9 hours (4) 9 minutes

Space for Rough Work

128. Duplicated DNA can be seen in all of the given phases of cell cycle in a cell undergoing cell division, **except**

(1) G_2 (2) G_1

(3) S (4) M

129. Which of the following statement is **Incorrect**?

(1) During mitosis, the main function of the spindle fibres is to repair the damaged DNA

(2) During leptotene, chromosomes look like thin threads

(3) Doubling of the number of chromosomes can be achieved by disrupting mitotic cell division soon after metaphase

(4) The proposition of Rudolf Virchow in cell theory was supported because of the occurrence of cell division

130. Terminalisation of chiasmata can be observed during

(1) Diakinesis (2) Diplotene

(3) Zygotene (4) Pachytene

131. If there are 24 centromeres in a cell during mitotic telophase at each pole, then how many number of chromosomes will be found in each daughter cell after cytokinesis?

(1) 12 (2) 24

(3) 48 (4) 6

132. Select the **correct** statement.

(1) Diptera has a greater number of common characters than Insecta

(2) Chordata has lesser number of organisms/members than Mammalia

(3) Muscidae and Dicotyledonae shares similar rank in taxonomical hierarchy

(4) Anacardiaceae share less number of common characters than Sapindales

133. Match Column-I with Column-II and select the **correct** option.

	Column-I		Column-II
I.	G_1 phase	a.	Duplication of polymer of deoxyribonucleotides
II.	S phase	b.	Synthesis of monomeric units of cytoskeletal element that form spindle fibre
III.	G_2 phase	c.	Synthesis of protein, RNA, ATP except DNA resulting in cell growth
IV.	M phase	d.	Helps in restoration of nucleo-cytoplasmic ratio after cell growth

	I	II	III	IV
(1)	a	d	c	b
(2)	b	a	d	c
(3)	c	b	a	d
(4)	c	a	b	d

134. Observe the following flow diagram:

Species → Genus → A → B → C → D → Kingdom

Select the statements that are the **correct** descriptions of the labels.

(1) A – These are characterised on the basis of both vegetative and reproductive features of plant species

(2) B – Similar characters are more in number as compared to different genera included in a family

(3) C – Include order mammalia along with order carnivora

(4) D – Represented by division chordata whose members are characterised by presence of notochord and dorsal hollow neural system

Space for Rough Work

135. Observe the following events:

(a) Splitting of primary constriction and formation of interzonal fibres

(b) Reformation of loose and indistinct network of nucleoprotein fibres

(c) Centrosome move towards pole

(d) Formation of phragmoplast

(e) Condensation of nucleoprotein fibres gets completed and shows congression

Which of the following option describes the sequential order of these events in a cell undergoing mitosis?

(1) b → d → a → c → e (2) d → c → a → e → b

(3) c → e → a → b → d (4) d → b → a → c → e

[ZOOLOGY]

136. The value of [redacted] in an adult man under normal conditions is almost equal to the

(1) Value of inspiratory capacity

(2) Two times to the value of IRV

(3) Equal to the residual volume

(4) Almost three times to the tidal volume

137. Which of the following gases is utilized by the organisms to breakdown [redacted] molecules?

(1) O_2 directly (2) CO_2 directly

(3) O_2 indirectly (4) CO_2 indirectly

138. The rate of diffusion of different gases across diffusion membrane of lungs does not depend upon

(1) Pressure/concentration gradient of a diffusible gas

(2) Solubility of diffusible gas

(3) Thickness of the diffusion membrane

(4) Total pressure of gaseous mixture in an inspired air

139. All of the following conducting parts of human's respiratory system are not supported by incomplete cartilaginous rings, **except**

(1) Initial bronchiole (2) Nasal passage

(3) Pharynx (4) Terminal bronchiole

140. **Assertion (A):** The amount of CO_2 that can diffuse through the diffusion membrane per unit difference in partial pressure is much [redacted] compared to that of O_2.

Reason (R): The solubility of CO_2 is 20-25 times [redacted] than that of O_2.

In the light of above statements, select the correct answer.

(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(2) Both (A) and (R) are true and (R) is the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

141. A healthy individual has [redacted] of haemoglobin in

(1) 5.5 millions of RBCs (2) 55 millions of RBCs

(3) Each litre of blood (4) Each decilitre of blood

142. Choose the correct option to complete the analogy w.r.t. human's blood.

[redacted] agranulocytes : Lymphocytes : : Most abundant formed elements : _____

(1) Neutrophils (2) Platelets

(3) Erythrocytes (4) Monocytes

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143. The condition erythroblastosis foetalis can be avoided in second pregnancy by administering
(1) Rh antibodies to the foetus just after parturition
(2) Rh antibodies to the mother during first trimester of pregnancy
(3) Anti-Rh antibodies to the foetus during second trimester of pregnancy
(4) Anti-Rh antibodies to Rh–ve mother immediately after the delivery of first Rh +ve child

144. Choose the **incorrect** statement w.r.t. tissue fluid.
(1) It is devoid of certain large proteins which are present in blood plasma.
(2) It has the same mineral distribution as that in blood plasma.
(3) It is important carrier for nutrients but not for hormones.
(4) It consists of water, water soluble substances, and lymphocytes.

145. Choose the correct statement w.r.t. coagulation of blood.
(1) Mechanism of coagulation is activated by certain factors released by stimulated platelets.
(2) Thrombokinase is formed by a cascade process that leads to its conversion into thrombin.
(3) Fibrinogens are responsible for clotting when blood with A antigen is transfused to individuals having anti-B antibodies.
(4) Clot is formed mainly of a network of threads called fibrinogens in which formed elements are trapped.

146. With the help of additional muscles in the abdomen, we have the ability to increase the strength of
(1) Normal inspiration only
(2) Forceful inspiration only
(3) Both forceful inspiration and forceful expiration
(4) Forceful expiration only

147. The factors present in our body which facilitate diffusion of O_2 from the alveoli to the deoxygenated blood, also facilitate diffusion of
(1) CO_2 from oxygenated blood to tissues
(2) O_2 from tissues to oxygenated blood
(3) CO_2 from alveoli to oxygenated blood
(4) O_2 from oxygenated blood to tissue fluid

148. All of the following are functions of conducting part of respiratory system, **except**
(1) Air conditioning of inspired air
(2) Providing surface for exchange of respiratory gases between alveolar air and blood
(3) Clearing of inhaled air from foreign particles
(4) Humidification of inhaled air

149. In humans, the volume of O_2 delivered to tissues from the blood pumped out by [redacted] under normal physiological conditions is about
(1) 500 mL (2) 250 mL
(3) 1250 mL (4) 200 mL

150. In humans, [redacted] involves following steps:
(a) Lifts up the ribs and sternum.
(b) Volume of thoracic chamber increases.
(c) Contraction of diaphragm and external inter-costal muscles.
(d) Air rushes into lungs due to suction pull
(e) Intra-pulmonary pressure decreases than atmospheric pressure

Choose the option that correctly represents the above given steps in a correct sequence.
(1) (c), (a), (b), (e), (d) (2) (a), (b), (c), (d), (e)
(3) (a), (b), (d), (c), (e) (4) (a), (d), (e), (b), (c)

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151. The evolution of animals with [redacted] circulatory system and a heart to pump the blood appeared for the first time in

(1) Helminths

(2) Annelids

(3) Arthropods

(4) Molluscs

152. [redacted] are best defined as blood vessels which

(1) Break up into capillaries that reunite to form an arteriole

(2) Carry deoxygenated blood from one visceral organ to another visceral organ

(3) Supply only oxygenated blood to the different organs

(4) Carry blood away from the heart to different organs of the body

153. A student catches two animals 'A' and 'B' from a pond. In animal 'A', deoxygenated blood is received by its atrium which gets oxygenated in gills. In animal 'B', oxygenated blood is collected from skin and lungs in left atrium and mixed blood is pumped by ventricle to the body tissues. Which type of circulation is present in animals 'A' and 'B' respectively?

(1) Single circulation; complete double circulation

(2) Incomplete double circulation; complete double circulation

(3) Single circulation; single circulation

(4) Single circulation; incomplete double circulation

154. Read the following statements carefully w.r.t. humans under normal conditions and choose the **correct** option.

(a) Ratio of diastolic and systolic blood pressure is equal to the ratio of pO_2 in blood at which haemoglobin is 75% saturated with O_2 and pO_2 in blood at which haemoglobin is 90% saturated with oxygen.

(b) Per cent of lymphocytes in the total WBCs is equal to the per cent of CO_2 transported in the form of carbamino-haemoglobin.

(c) Per cent of total CO_2 transported through blood plasma as bicarbonate is [redacted] to the per cent of CO_2 transported in a dissolved state *via* the blood plasma.

(1) All the statements (a), (b) and (c) are incorrect

(2) Only statements (a) and (b) are correct

(3) Only statement (a) is incorrect

(4) All the statements (a), (b) and (c) are correct

155. Which of the following statements is correct?

(1) Blood moves freely from vena cava to ventricles through atria during joint diastole.

(2) Closure of semilunar valves occurs during ventricular systole.

(3) Closure of AV valves occurs during atrial systole.

(4) Both AV valves and semilunar valves of heart remain open during complete duration of joint diastole.

156. A patient is suffering from bradycardia and his heart rate is about 50 times per minute. If the end diastolic volume of each ventricle is [redacted] and end systolic volume is 60 mL, what will be the amount of blood pumped by left ventricle into the aorta?

(1) 4 Litres/minute (2) 5 Litres/minute

(3) 3 Litres/minute (4) 6 Litres/minute

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157. Match the following columns and select the **correct** option.

	Column I		Column II
A.	Coronary system of blood vessels	(I)	A link between arteriole and venule
B.	Hepatic portal vein	(II)	Supply blood to heart musculature
C.	Blood capillary	(III)	Opens into right atrium of heart
D.	Vena cava	(IV)	Drains blood from intestine to liver

(1) A-(I), B-(II), C-(III), D-(IV)

(2) A-(II), B-(IV), C-(I), D-(III)

(3) A-(IV), B-(III), C-(II), D-(I)

(4) A-(IV), B-(II), C-(I), D-(III)

158. Read the following statements carefully.

Statement (A): During T-P gap of a standard ECG, heart remains in a electrically silent stage.

Statement (B): [redacted] release a destructive enzyme called histaminase that increases phagocytosis of eosinophils.

Select the correct option.

(1) Statement (A) is correct but statement (B) is incorrect

(2) Statement (A) is incorrect but statement (B) is correct

(3) Both the statements (A) and (B) are correct

(4) Both the statements (A) and (B) are incorrect

159. Select the layer(s) of blood vessels which have/has two types of fibres, each having property of elasticity.

(1) Tunica intima

(2) Tunica externa

(3) Tunica media

(4) Tunica externa and tunica intima

160. Choose the **incorrect** match.

(1)	Neutrophils	–	Phagocytosis
(2)	Basophils	–	Anti-inflammatory reactions
(3)	Eosinophils	–	Resist infections
(4)	Lymphocytes	–	Immune responses

161. Select the correct option to complete the analogy.

SAN : Right upper corner of right atrium : : [redacted] : _____

(1) Right upper corner of right atrium

(2) Right upper corner of left atrium

(3) Lower left corner of left atrium

(4) Lower left corner of right atrium

162. Which of the following can [redacted] the rate of heart beat, the strength of ventricular contraction and the cardiac output?

(a) A special neural centre in pons

(b) Neural signals from sympathetic neural system

(c) Neural signals from parasympathetic neural system

(d) Hormones from thymus

(e) Hormones from adrenal medulla

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Choose the correct option.

(1) (a), (b) and (c) only (2) (b), (c) and (d) only

(3) (c), (d) and (e) only (4) (b) and (e) only

163. Which of the following features/properties of mammalian RBCs make them more efficient in their function of transporting oxygen?

(1) Presence of haemoglobin exhibiting tertiary structure

(2) Presence of enzyme carbonic anhydrase with Ca^{2+} as the cofactor

(3) Biconcave shape and absence of nucleus

(4) Permeability of their plasma membrane

164. Select the **incorrect** statement.

(1) Reduction in number of platelets leads to clotting disorder.

(2) Serum differs from blood plasma in lacking clotting factors.

(3) Blood pressure is maximum in aorta and minimum in capillaries during ventricular systole.

(4) Person with blood group AB–ve is considered as universal recipient.

165. The nodal tissue which has slowest conduction velocity of impulses and delays spreading of impulses so that atria and ventricles contract separately in a synchronised fashion is 'X'.

The nodal tissue which connects atria to ventricle so that impulses from atrial nodal tissues reach into ventricles is 'Y'.

Choose the option that correctly identifies 'X' and 'Y' respectively.

(1) SA node, AV Bundle

(2) AV node, Purkinje fibres

(3) AV node, AV Bundle

(4) SA node, Bundle branches

166. During normal [redacted] in humans, there is an [redacted] in

(a) Intra-abdominal pressure

(b) Intra-pulmonary volume

(c) Intra-pleural pressure

(d) Length of thoracic cavity

Choose the most appropriate option.

(1) (a), (b) and (c) (2) (b), (c) and (d)

(3) (a), (c) and (d) (4) (a), (b) and (d)

167. Variations in which of the following components of body fluids do not affect respiration?

(1) HCO_3^- concentration in arterial oxygenated blood

(2) Na^+ concentration in arterial deoxygenated blood

(3) CO_2 concentration in cerebrospinal fluid

(4) H^+ concentration in arterial oxygenated blood

168. Which of the following conditions would result in the [redacted] of saturation of haemoglobin with oxygen, assuming pO_2 remains constant?

(1) Increase in CO_2 level, decrease in temperature

(2) Decrease in CO_2 level, decrease in temperature

(3) Increase in CO_2 level, increase in H^+ concentration

(4) Decrease in CO_2 level, decrease in pH

169. The [redacted] of human thoracic cavity are respectively formed by

(1) Diaphragm; sternum

(2) Vertebral column; ribs

(3) Diaphragm; ribs

(4) Ribs; diaphragm

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170. Match List-I with List-II w.r.t. humans.

	List-I		List-II
(A)	Asthma	(I)	Decreased respiratory surface
(B)	Emphysema	(II)	Lung fibrosis
(C)	Occupational respiratory disorder	(III)	Inflammation of bronchi and bronchioles
(D)	Coronary artery disease	(IV)	Lipid rich diet

Choose the correct option.

(1) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

(2) (A)-(III), (B)-(I), (C)-(II), (D)-(IV)

(3) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

(4) (A)-(II), (B)-(IV), (C)-(III), (D)-(I)

171. Read the following statements carefully and choose the **correct** option.

Statement (A): Injection of a drug that stimulates the chemoreceptors in carotid bodies would be expected to cause an [redacted] in pO_2 of arterial blood.

Statement (B): O_2 delivery to tissues would be [redacted] to a large extent in a patient with carbon monoxide poisoning.

(1) Both statements (A) and (B) are correct

(2) Both statements (A) and (B) are incorrect

(3) Only statement (A) is correct

(4) Only statement (B) is correct

172. Which of the following animals circulate water from their surroundings through their body cavities to facilitate the cells to exchange substances?

(1) Annelids and flatworms

(2) Arthropods and echinoderms

(3) Poriferans and coelenterates

(4) Echinoderms and hemichordates

173. Select the **incorrect** match w.r.t. human blood.

(1) Formed elements — 45% of the total blood

(2) Neutrophils — 60-65% of the total blood cells

(3) Eosinophils — 2-3% of the total WBCs

(4) Blood plasma — 55% of the total blood

174. Read the following statements w.r.t. a standard ECG:

(a) To obtain a standard ECG, multiple leads are attached to the chest region.

(b) Each peak in ECG is identified with a letter from P to T that corresponds to a special mechanical activity of the heart.

(c) Contraction of ventricles starts slightly after Q-wave and marks the beginning of ventricular systole.

(d) One can determine the heart beat rate of an individual by counting the number of QRS complexes that occur in a given time period.

Choose the option with only [redacted] statements.

(1) (a), (b) and (d) only

(2) (b) and (c) only

(3) (a) and (b) only

(4) (a), (c) and (d) only

175. Choose the **Incorrect** match w.r.t. an adult human.

(1) Respiratory minute volume — 6000-8000 mL

(2) Inspiratory capacity — 3000-3500 mL

(3) Vital capacity — 2500-3000 mL

(4) Residual volume — 1100-1200 mL

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176. After forceful expiration, the residual volume of air remains in lungs and cannot be expelled out, because
(1) Intra-pulmonary pressure is more than atmospheric pressure on lung walls
(2) There is a negative intra-pleural pressure which prevents collapsing of lung alveoli
(3) There is a positive intra-pleural pressure pulling on the wall of lungs
(4) Negative intra-pulmonary pressure than atmospheric pressure

177. P_{50} value is defined as the partial pressure of oxygen at which haemoglobin is fifty per cent saturated with oxygen and its value is about
(1) 10-20 mmHg (2) 25-30 mmHg
(3) 40-50 mmHg (4) 50-60 mmHg

178. **Assertion (A):** [redacted] and left ventricle of heart are included in systemic circulation.
Reason (R): Right atrium receives deoxygenated blood from all the systems of the body and left ventricle pumps oxygenated blood to all systems of the body.
In the light of above statements, select the correct answer from the options given below.
(1) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(2) Both (A) and (R) are true and (R) is the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

179. Choose the **Incorrect** match.

(1)	Angina pectoris	–	Acute chest pain
(2)	Heart failure	–	Heart stops beating
(3)	Heart attack	–	Heart muscle is suddenly damaged
(4)	Hypertension	–	Blood pressure > 140/90 or more on repeated checks

180. Choose the correct statement w.r.t. cardiac cycle in humans.
(1) Complete filling of ventricles with blood takes place during joint diastole.
(2) Heart sound 'Lub' is produced due to closure of semilunar valves during ventricular systole.
(3) Opening of AV valves occurs due to atrial systole.
(4) 'Dub' sound of heart is produced during ventricular diastole.

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